EXPVID: A BENCHMARK FOR EXPERIMENT VIDEO UNDERSTANDING & REASONING

Anonymous Authors

ABSTRACT

ExpVid is a benchmark for evaluating multimodal large language models on authentic wet-lab experiment videos. Existing video benchmarks emphasize generic activities, narrative long-video reasoning, or domain knowledge questions, but they rarely test whether a model can track fine-grained laboratory actions, recover procedural structure, and connect the observed procedure to scientific conclusions in a visually grounded way. We construct ExpVid from 390 peer-reviewed JoVE experiment videos spanning 13 disciplines, with an average duration of 489 seconds, and annotate 7,800 question-answer pairs organized into a three-level hierarchy: fine-grained perception, procedural understanding, and scientific reasoning. The benchmark covers 10 tasks, ranging from recognizing tools, materials, quantities, and operations, to step ordering, completeness verification, and cloze-style analysis and discovery questions over full experiments. Its construction combines LLM-assisted preprocessing and drafting with PhD-level expert verification, while deliberately removing obvious textual shortcuts in the reasoning level. Across 19 model families and 22 evaluated configurations, ExpVid reveals a consistent pattern: current models can often recognize local actions and reorder known steps, but they remain substantially weaker at sequence generation, next-step prediction, and experiment-level reasoning. The strongest closed-source systems lead overall, with GPT-5 reaching 57.5 on Level-2 and 56.4 on Level-3, while the best Level-1 score is 60.2 from Gemini-2.5-Flash in thinking mode. Our analyses further show that visual input is indispensable and that extra test-time reasoning is not uniformly beneficial. These results position ExpVid as a benchmark for measuring what current video MLLMs understand about experimental procedure, and where they still fail.

1 Introduction

Multimodal large language models (MLLMs) are increasingly evaluated on video understanding benchmarks such as MVBench, Video-MME, MLVU, LVBench, and VRBench, which probe temporal perception, long-context comprehension, and multi-step reasoning in broad real-world or narrative settings [Li et al., 2024, Fu et al., 2025, Zhou et al., 2024, Wang et al., 2024c, Yu et al., 2025]. In parallel, expert-oriented multimodal benchmarks such as MMMU, MMVU, Video-MMMU, MathVista, MathVision, MMStar, SciBench, and SFE assess knowledge-intensive reasoning in images, documents, and professional videos [Yue et al., 2024, Zhao et al., 2025, Hu et al., 2025, Lu et al., 2023, Wang et al., 2024b, Chen et al., 2024, Wang et al., 2023, Zhou et al., 2025]. However, these evaluations leave a specific gap: they do not directly measure whether a model can watch a real laboratory experiment, ground its answers in the visual evidence, reconstruct procedural structure over minutes, and reason from the observed protocol to scientifically meaningful conclusions.

This gap matters because laboratory videos are neither generic web videos nor textbook science questions. Wet-lab experiments are long-horizon, visually dense, and procedurally constrained. Success requires identifying fine-grained tools and materials, distinguishing subtle operations, tracking what has and has not been executed, and connecting the observed procedure to the aims and findings of the experiment. Prior work on biological experiment video analysis has focused mainly on activity recognition and structured annotation rather than MLLM evaluation [Yagi et al., 2025, Gabrieli et al., 2025]. As a result, current benchmark suites provide limited evidence about whether frontier MLLMs can support experiment-centric scientific workflows.

We introduce ExpVid, a benchmark for experiment video understanding and reasoning built from peer-reviewed videos from the Journal of Visualized Experiments (JoVE). ExpVid is organized as a three-level hierarchy that mirrors the scientific workflow: Level-1 tests fine-grained perception over short clips, Level-2 tests procedural understanding over stage-level segments, and Level-3 tests scientific reasoning over full experiments. The benchmark contains 7,800

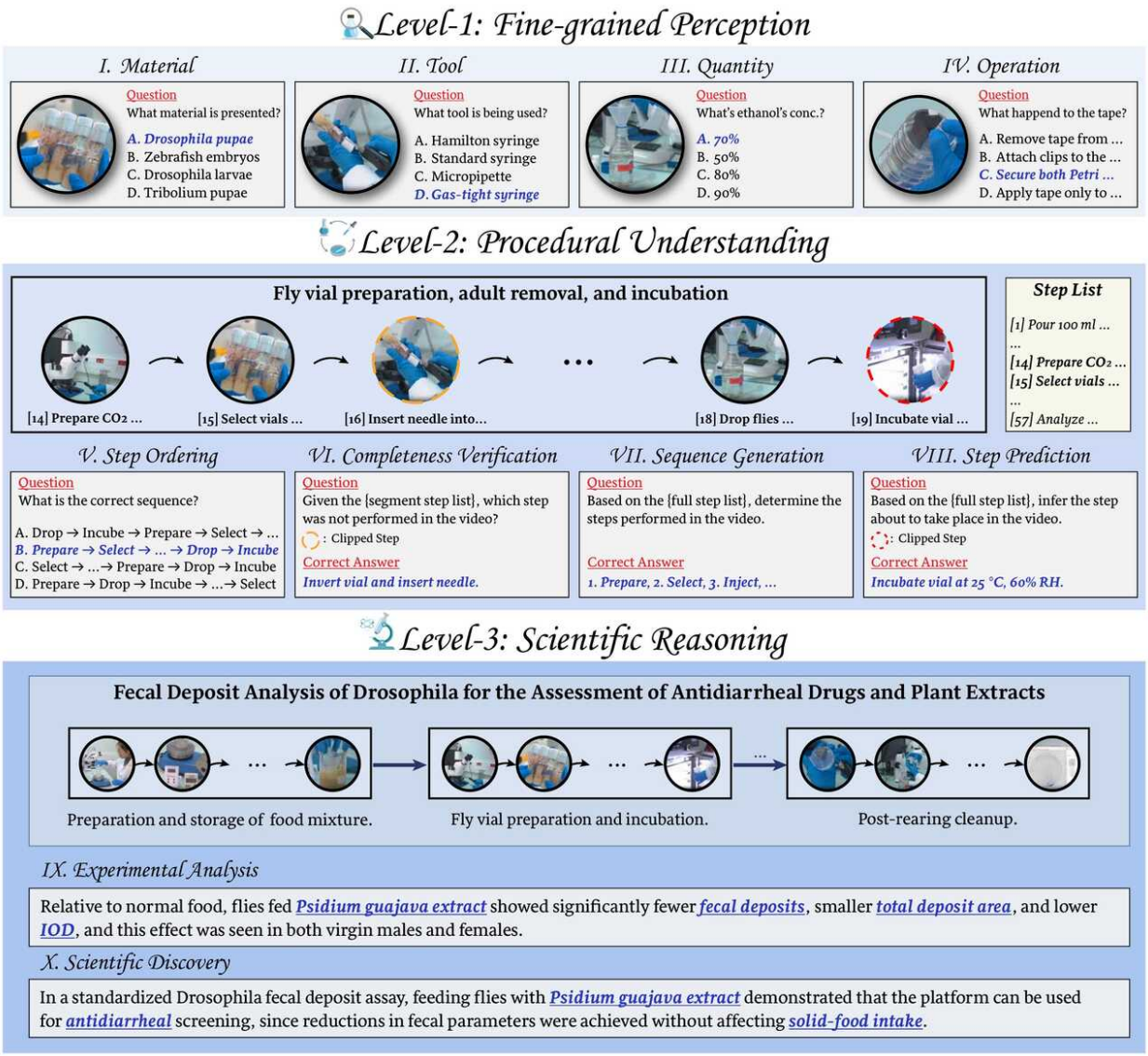

Figure 1: Overview of the three-level task hierarchy in ExpVid. Level-1 focuses on fine-grained perception, Level-2 on procedural understanding, and Level-3 on experiment-level scientific reasoning.

verified question-answer pairs across 10 tasks, derived from 390 videos spanning 13 disciplines and multiple temporal scales. The construction pipeline uses LLMs for filtering, segmentation, and draft annotation, but keeps the benchmark vision-centric through visibility checks, expert verification, and the removal of obvious textual shortcuts.

Our empirical study evaluates 19 model families under 22 configurations, including open- and closed-source MLLMs with and without explicit thinking modes. The main findings are consistent across levels. First, closed-source systems currently lead overall, but performance remains far from saturated. Second, procedural understanding is uneven: models are strong at step ordering yet weak at sequence generation and especially next-step prediction. Third, additional test-time reasoning is unreliable, sometimes improving perception while slightly hurting procedural or experiment-level reasoning. Fourth, video frames are essential: removing them causes large degradations, especially for Levels 2 and 3.

The contributions of this paper are threefold. We introduce a benchmark for authentic experiment video evaluation that spans perception, procedure, and scientific reasoning. We present a semi-automatic but expert-verified construction pipeline designed to reduce text shortcuts and preserve visual grounding. We provide a broad empirical study showing what current MLLMs can and cannot infer from real wet-lab videos.

Table 1: Comparison between ExpVid and some MLLM benchmarks. A and M indicate automatic and manual annotation, respectively.

Benchmark	#QA Pairs	#Videos	Avg. Sec.	#Task Types	Annotation	Domain
General Video Benchmarks						
MVBench [Li et al., 2024]	4,000	3,641	16.0	20	A+M	General
Video-MME [Fu et al., 2025]	2,700	900	1,017.9	1	M	General
MLVU [Zhou et al., 2024]	3,102	1,730	930.0	9	M	Narrative
VRBench [Yu et al., 2025]	9,468	960	5,796.0	1	M	Narrative
Knowledge-driven Benchmarks						
MMVU [Zhao et al., 2025]	3,000	1,529	51.4	2	M	Multi-disc.
Video-MMMU [Hu et al., 2025]	900	300	506.2	3	M	Multi-disc.
MathVision [Wang et al., 2024b]	3,040	–	–	1	M	Math
MathVista [Lu et al., 2023]	6,141	–	–	31	M	Math
MMMU [Yue et al., 2024]	11,500	–	–	2	M	Multi-disc.
ScienceQA [Saikh et al., 2022]	21,208	–	–	1	M	Science
SciBench [Wang et al., 2023]	789	–	–	1	M	Science
MMStar [Chen et al., 2024]	1,500	–	–	6	M	Multi-disc.
SFE [Zhou et al., 2025]	830	–	–	66	M	Science
ExpVid	7,800	390	489.0	10	A+M	Science

2 Related work

Existing work relevant to ExpVid falls into three groups.

General video understanding benchmarks. Recent MLLM benchmarks substantially improved evaluation for temporal and long-form video understanding. MVBench organizes diverse temporal perception tasks over short clips [Li et al., 2024]. Video-MME broadens video duration and modality coverage [Fu et al., 2025]. MLVU and LVBench emphasize long-video understanding [Zhou et al., 2024, Wang et al., 2024c], while EgoSchema, E.T. Bench, and VRBench stress event-level and multi-step reasoning in egocentric or narrative settings [Mangalam et al., 2023, Liu et al., 2024, Yu et al., 2025]. These benchmarks are important, but they are not centered on scientific experiments, where the main challenge is to recover protocol structure and draw scientific conclusions from visually grounded laboratory procedures.

Scientific and knowledge-intensive multimodal benchmarks. MMMU, MathVista, MathVision, MMStar, SciBench, and SFE evaluate expert-level multimodal or scientific reasoning, often with diagrams, textbook figures, charts, or curated visual questions [Yue et al., 2024, Lu et al., 2023, Wang et al., 2024b, Chen et al., 2024, Wang et al., 2023, Zhou et al., 2025]. MMVU and Video-MMMU move into professional and video-based settings [Zhao et al., 2025, Hu et al., 2025]. These benchmarks show that domain knowledge remains challenging, but their videos are not authentic wet-lab procedures, and they do not explicitly separate fine-grained laboratory perception from procedure tracking and experiment-level inference.

Laboratory video datasets and scientific activity analysis. Datasets such as FineBio and recent activity-recognition work on scientific experimentation target hierarchical action annotation or recognition in laboratory settings [Yagi et al., 2025, Gabrieli et al., 2025]. This line of work is complementary to ours: it studies structured action understanding in scientific environments, whereas ExpVid is an evaluation benchmark for MLLMs that uses authentic, peer-reviewed experiment videos and measures not only actions but also procedural fidelity and reasoning.

Table 1 places ExpVid among representative video and science-oriented benchmarks. Relative to general video benchmarks, ExpVid is narrower in domain but more specialized in task design. Relative to science benchmarks, it contributes authentic long-form videos and visually grounded procedure-centered evaluation.

3 Benchmark Design

ExpVid is designed to evaluate three progressively harder capabilities required for understanding experiment videos: local perception, procedure reconstruction, and scientific reasoning. The benchmark uses video as the primary ev-

idence source at evaluation time; paired transcripts and papers are used only during construction. This design is intended to reduce shortcutting from text priors and to make failure cases more diagnostic.

3.1 Source collection and scope

We source videos from JoVE Research, a collection of peer-reviewed laboratory videos paired with structured articles. Starting from roughly 15K crawled candidates, we retain videos with associated ASR transcripts and papers, then filter them using an LLM-based scoring stage guided by continuity, alignment, clarity, integrity, and focus. Following the provided protocol, videos are further restricted to the interquartile duration range of 378–728 seconds, and experts curate a final set of 390 experiments, with 30 videos from each of 13 disciplines. The resulting benchmark covers five broad areas shown in Figure 2: biological sciences, medicine and health, neuroscience, engineering, and environmental science.

3.2 Three-level task hierarchy

Figure 1 summarizes the task design.

Level-1: Fine-grained perception. Level-1 operates on short clips derived from sentence-aligned video segments, with an average duration of about 8 seconds. It contains four multiple-choice tasks: *Material Recognition*, *Tool Recognition*, *Quantity Recognition*, and *Operation Recognition*. These tasks test whether a model can localize the visually relevant evidence and distinguish confusable laboratory entities and actions.

Level-2: Procedural understanding. Level-2 uses stage-level segments with an average duration of about 48 seconds. It contains four tasks: *Step Ordering*, *Sequence Generation*, *Completeness Verification*, and *Step Prediction*. These tasks move beyond local recognition and test whether a model can infer which steps occurred, whether a segment is missing a required step, and what should happen next according to the protocol.

Level-3: Scientific reasoning. Level-3 uses the full experiment video, with average duration close to 8 minutes. It contains two cloze-style tasks: *Experimental Analysis* and *Scientific Discovery*. These questions require integrating the observed protocol with the paired paper to infer experiment-specific conclusions, while preventing answerability from the article text alone.

3.3 Annotation pipeline

The construction workflow is shown in Figure 2. The pipeline is semi-automatic, but every final question is expert verified.

Preprocessing. For Level-1, ASR sentences are used to segment videos into about 10K short clips. For Level-2, an LLM-based stage segmentation procedure identifies 20–60 second segments and extracts step lists, producing roughly 3.5K stage-level clips. For Level-3, the full procedure is retained, but concluding slides and explicit data-analysis sections are removed to reduce shortcuts from narration or result slides.

LLM-assisted drafting. The pipeline uses DeepSeek-R1 for filtering and step extraction [Guo et al., 2025a], Qwen2.5-VL for visibility checking and caption-based support [Bai et al., 2025b], MinerU for extracting relevant paper sections [Wang et al., 2024a], and GPT-5 for anchor summarization and question drafting [OpenAI, 2025]. Level-1 distractors are constructed to be visually or functionally plausible. Quantity questions use nearby numeric distractors, and operation questions use confusable actions. Level-2 questions are built from extracted step lists and controlled perturbations of those lists. Level-3 questions are written in cloze form around experimentally meaningful conclusions rather than raw textual recall.

Expert verification. PhD-level annotators in biology, chemistry, and medicine validate the visibility of the target evidence, the plausibility of distractors, alignment between steps and segments, and the uniqueness of the intended answer. According to the provided annotation notes, the average review time is approximately 0.3 hours per Level-1 item, 0.5 hours per Level-2 item, and 1.2 hours per Level-3 item. The final benchmark contains 7,800 verified QA pairs. The task distribution is moderately balanced: the largest slice is Material Recognition (16.2%), while each Level-3 task contributes 5.0% of the benchmark, as shown in Figure 2.

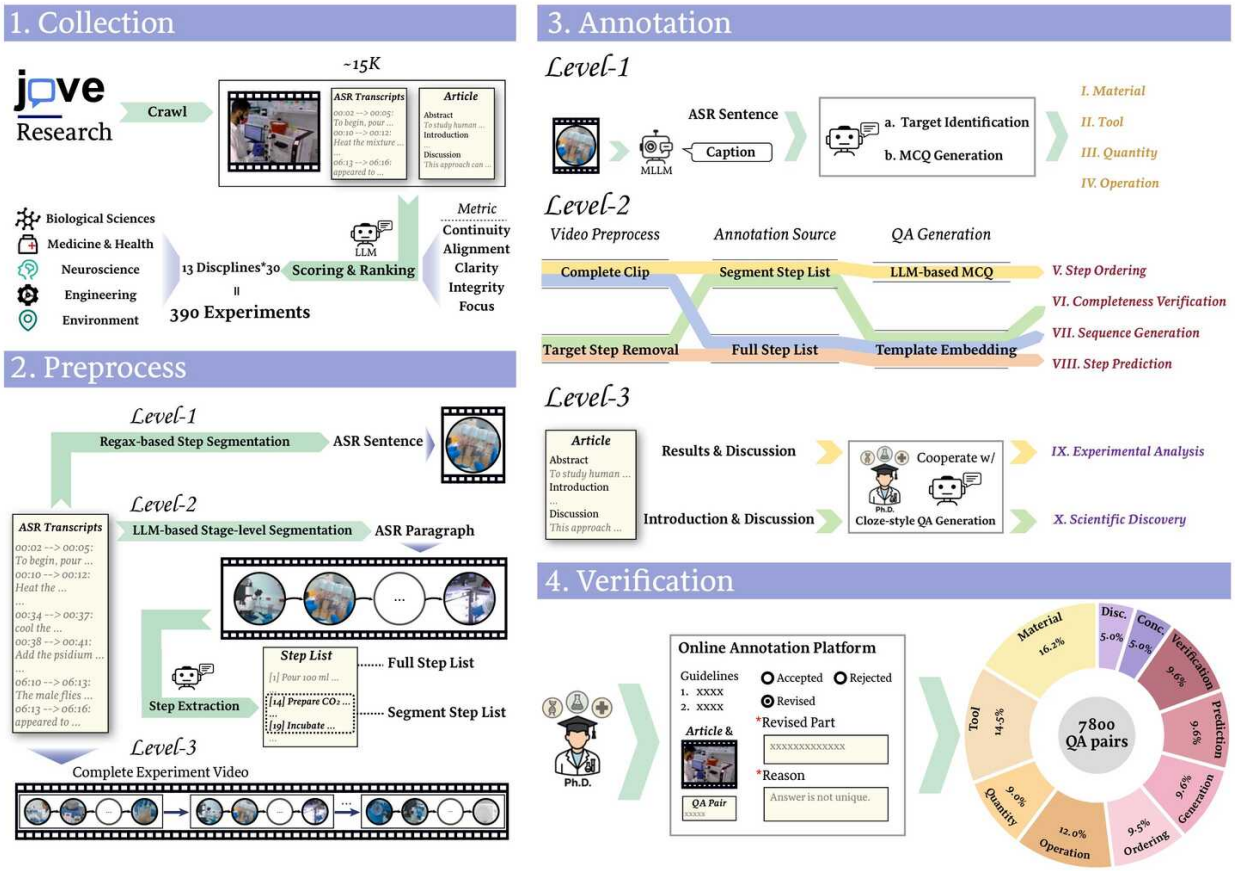

Figure 2: Construction and verification pipeline for ExpVid. The benchmark is built from JoVE experiment videos, ASR transcripts, and paired papers, then organized into three temporal levels with LLM-assisted drafting and expert verification.

3.4 Evaluation protocol

For multiple-choice tasks, we report Top-1 accuracy. For *Sequence Generation*, we evaluate the overlap between the predicted performed-step set and the gold set using Jaccard similarity:

$$\mathrm{Jaccard}(\hat{S}, S) = \frac{|\hat{S} \cap S|}{|\hat{S} \cup S|}.$$

For Level-3 cloze questions, we report per-blank accuracy using an LLM judge, following the provided protocol, to allow semantically equivalent answers that are not string-identical. We report human performance when available; the released results include human baselines for Levels 1 and 2 only.

4 Benchmark Validation

Because ExpVid is intended as an evaluation instrument rather than a dataset-only release, validating what it measures is central.

Visual dependency. ExpVid is designed to be vision-centric. Level-1 questions require the target tool, material, quantity cue, or operation to be visible in the clip. Level-2 questions are anchored to video segments rather than transcript snippets. Level-3 removes post-hoc conclusion slides and explicit analysis sections that would otherwise let models answer from text alone. The frame-ablation study in Figure 3 supports this design: for the Gemini-2.5-Flash configuration shown, removing frames drops Level-1 average from 58.6 to 39.6 and Level-2 average from 50.1 to 28.3, with especially large losses on sequence generation and step prediction.

Procedural validity. The benchmark separates short-clip recognition from stage-level and full-experiment reasoning. This matters because good local perception does not imply procedural understanding. The strongest ordering scores exceed 85, but next-step prediction remains below 52 even for GPT-5 and near zero for some otherwise capable models. This discrepancy indicates that the tasks are not redundant with one another and that the benchmark distinguishes memorizing local cues from modeling procedure.

Difficulty calibration. Table 2 further suggests that ExpVid is challenging without collapsing into either ceiling effects or universal failure. Some slices are clearly accessible: several models exceed 70 on quantity recognition, the best operation score reaches 80.8, and human annotators score 69.8 on step ordering. Yet other tasks remain far from saturated, including tool and material recognition, completeness verification, and especially next-step prediction. The benchmark therefore spans a useful range of difficulty across temporal scales. This spread is desirable because it allows strong systems to separate on easier perceptual tasks while still leaving substantial headroom on procedure-centric reasoning, where even GPT-5 reaches only 51.8 on prediction and 57.5 on the Level-2 average.

Expert review and quality control. Every final item is reviewed by domain experts, and the pipeline explicitly checks visibility, answer uniqueness, and step-segment alignment. These checks help avoid the two common failure modes documented in multimodal evaluation: questions answerable without the visual input and hidden ambiguity in the annotation [Chen et al., 2024]. At the same time, we do not report inter-annotator agreement statistics, so expert validation should be interpreted as a strong but not exhaustive quality signal.

Scope boundaries. ExpVid measures video-grounded understanding of wet-lab procedures and related conclusions. It does not measure physical execution ability, open-ended scientific creativity, or general laboratory safety competence. Since JoVE videos and papers are public, ExpVid also should not be treated as contamination-proof.

5 Experimental Setup

We evaluate 19 model families under 22 total configurations, covering both open-source and closed-source MLLMs. The evaluated open-source families include Qwen2.5-VL, MiMo-VL, InternVL3.5, Intern-S1, Keye-VL, GLM-4.xV, and Kimi-VL [Bai et al., 2025b, Wang et al., 2025, Bai et al., 2025a, Hong et al., 2025, Team et al., 2025]. Closed-source systems include Seed-VL-1.5, Claude Sonnet 4, Gemini 2.5 Flash, Gemini 2.5 Pro, and GPT-5 [Guo et al., 2025b, Anthropic, 2025, DeepMind, 2025a,b, OpenAI, 2025]. When a model exposes both standard and explicit "thinking" inference modes, we report both settings.

All models are evaluated on the benchmark tasks defined in Section 3 under the provided scoring protocol. Multiple-choice tasks use direct answer selection, sequence generation uses Jaccard overlap, and Level-3 cloze tasks use per-blank judged accuracy. Human baselines are reported for Levels 1 and 2 as part of the released results table.

6 Benchmark Results

Table 2 reports the main benchmark results.

Overall performance. Closed-source systems lead on most aggregates, but the benchmark remains unsolved. Gemini-2.5-Flash in thinking mode achieves the best Level-1 average at 60.2, while GPT-5 is best on Level-2 and Level-3 with 57.5 and 56.4, respectively. Among open-source systems, InternVL3-78B is strongest on Levels 1 and 2 with 50.9 and 41.9, whereas Intern-S1 attains the best open Level-3 score at 39.6. Notably, the best open Level-2 score remains slightly below the human baseline of 42.1.

Fine-grained perception is challenging but tractable. Level-1 is the strongest level for many models, especially on quantities and operations. The best quantity score is 73.2 from InternVL3-78B, and the best operation score is 80.8 from Gemini-2.5-Pro. In contrast, tool and material recognition remain difficult: even the best tool and material scores are only 53.1 and 50.7. This pattern suggests that visually subtle laboratory entities are a persistent bottleneck.

Procedural understanding is highly asymmetric. The most striking pattern appears in Level-2. Step Ordering is relatively strong across many systems, peaking at 87.1 for InternVL3-78B. However, Sequence Generation is much harder, with GPT-5 reaching 66.9 and most open models far lower. Completeness Verification remains below 27 for every model, and Step Prediction is the hardest task overall: GPT-5 reaches 51.8, Gemini-2.5-Pro 49.6, but several strong open models score close to zero, including Qwen2.5-VL-72B at 0.3. These gaps show that recovering an observed order is substantially easier than inferring latent omissions or anticipating the next protocol step.

Table 2: Performance of evaluated models on the ExpVid across 10 tasks under three levels.

Model	Think	Level-1					Level-2					Level-3		
		Tool	Mat.	Quan.	Oper.	Avg.	Ord.	Gen.	Veri.	Pred.	Avg.	Anal.	Disc.	Avg.
Human Performance		17.5	15.9	61.3	55.5	37.6	69.8	31.2	45.6	21.8	42.1	–	–	–
Open-source MLLMs														
Qwen2.5-VL-7B-Instruct	×	32.0	33.9	49.0	62.4	42.6	56.2	20.8	20.7	1.3	24.6	25.2	21.4	23.3
MiMo-VL-7B-RL	×	34.2	33.7	44.2	62.4	42.4	43.9	28.5	18.5	11.4	27.4	28.7	25.9	27.3
MiMo-VL-7B-RL	✓	36.1	29.1	53.6	67.8	44.3	64.8	32.3	24.9	15.6	34.3	29.3	27.3	28.3
InternVL3-8B	×	27.5	31.0	38.8	65.6	39.4	43.4	20.4	20.2	3.9	23.9	29.2	25.3	27.2
InternVL3.5-8B	×	27.3	30.8	45.5	64.8	40.3	82.3	25.8	23.7	4.8	34.0	22.6	18.4	20.5
Intern-S1-mini	✓	33.3	31.2	52.5	61.4	42.5	73.6	14.3	16.8	8.3	28.1	33.5	28.3	30.9
Keye-VL-8B-Preview	✓	16.6	22.4	38.9	60.8	32.6	25.4	12.4	19.1	1.7	14.6	9.5	6.7	8.1
Keye-VL-1.5-8B	✓	21.0	23.4	51.3	64.0	37.0	56.7	9.5	20.0	2.8	22.1	8.4	6.1	7.2
GLM-4.1V-9B	✓	30.8	29.8	47.5	59.6	40.1	64.1	18.2	25.0	7.4	28.6	28.1	26.5	27.3
GLM-4.5V	✓	35.5	33.6	61.5	62.3	45.6	71.9	34.9	27.2	12.9	36.6	33.3	32.5	32.9
Kimi-VL-A3B-Thinking	✓	34.6	32.6	40.7	59.5	40.8	32.3	18.2	23.3	6.2	20.0	24.6	21.8	23.2
InternVL3.5-38B	✓	35.9	34.0	46.7	65.3	44.0	65.8	36.7	23.0	19.0	36.0	33.1	30.8	31.9
InternVL3-78B	✓	35.1	34.3	**73.2**	75.8	50.9	**87.1**	45.5	19.8	15.5	41.9	40.3	35.3	37.7
Qwen2.5-VL-72B-Instruct	×	30.5	34.7	54.5	64.5	43.9	86.3	34.1	23.8	0.3	35.9	31.9	29.3	30.6
Intern-S1	✓	38.9	35.2	58.9	73.8	49.9	82.2	45.0	24.1	15.4	36.0	43.0	36.3	39.6
Closed-source MLLMs														
Seed-VL-1.5	✓	32.9	24.6	43.9	69.2	40.7	73.9	48.6	19.8	27.9	42.5	32.0	29.4	30.7
Claude-Sonnet-4	×	25.6	31.2	54.3	61.9	40.8	78.7	37.6	16.5	11.6	36.0	29.1	30.1	29.6
Gemini-2.5-Flash	×	52.7	50.1	65.2	72.6	58.6	86.0	50.5	24.1	40.2	50.1	47.2	41.1	44.1
Gemini-2.5-Flash	✓	52.7	**50.7**	71.9	73.3	**60.2**	85.1	54.3	22.3	38.0	49.8	44.8	41.3	43.0
Gemini-2.5-Pro	×	**53.1**	45.9	64.3	**80.8**	59.2	83.7	61.3	**26.8**	49.6	53.8	50.6	45.2	47.9
Gemini-2.5-Pro	✓	51.3	44.3	63.8	74.4	56.7	84.2	59.9	**26.8**	46.9	54.3	50.1	44.8	47.4
GPT-5	✓	51.6	37.8	59.5	71.9	53.3	85.1	**66.9**	**26.8**	**51.8**	**57.5**	**55.4**	**57.4**	**56.4**

Experiment-level reasoning remains open. On Level-3, GPT-5 leads both *Experimental Analysis* and *Scientific Discovery* with 55.4 and 57.4, respectively. The best open-source result is 43.0 on analysis and 36.3 on discovery from Intern-S1, with a 39.6 average. Even strong systems therefore leave a sizable gap on cloze-style reasoning over full experiments.

7 Analysis and Discussion

Thinking mode is not uniformly helpful. Explicit thinking improves some models but not others. MiMo-VL-7B-RL gains across all three levels when thinking is enabled, improving from 42.4 to 44.3 on Level-1 and from 27.3 to 28.3 on Level-3. In contrast, Gemini-2.5-Flash improves from 58.6 to 60.2 on Level-1 but slightly declines from 50.1 to 49.8 on Level-2 and from 44.1 to 43.0 on Level-3. Gemini-2.5-Pro shows a similar small regression under thinking mode. On this benchmark, longer reasoning traces do not reliably translate into better video-grounded answers and may sometimes drift away from the visual evidence.

Frames are essential, especially beyond Level-1. Figure 3 directly probes visual dependence. For the Gemini-2.5-Flash setup, removing frames sharply reduces performance on every Level-1 and Level-2 task, with especially severe degradation on Sequence Generation (50.5 to 12.5) and Step Prediction (40.2 to 5.8). This confirms that ExpVid is not mainly solved by language priors or option statistics. The right side of Figure 3 further shows that additional frames help long-horizon Level-3 reasoning for some models, but not monotonically for all of them. Kimi-VL and MiMo-VL continue to improve as the frame budget increases toward 256, whereas InternVL3.5-8B peaks earlier and

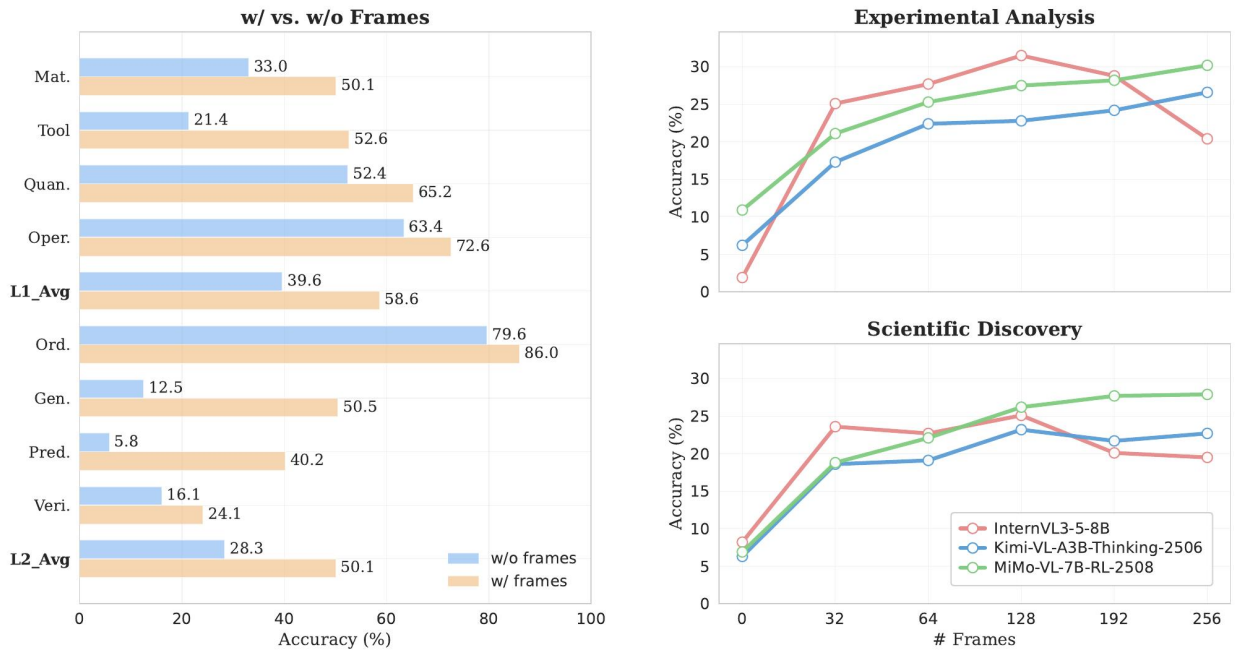

Figure 3: Effect of visual input in ExpVid. Left: removing frames causes large drops on Level-1 and Level-2 for the Gemini-2.5-Flash configuration shown. Right: Level-3 accuracy changes with the number of input frames for three open-source models, indicating that more visual context helps some long-context models but not uniformly.

then degrades. This suggests that long-context capacity and frame selection strategy materially affect experiment-level reasoning.

Cross-level rankings are unstable. A useful benchmark should change the conclusions one draws from a single aggregate score, and ExpVid does. The model that leads Level-1 is not the model that leads Levels 2 and 3: Gemini-2.5-Flash in thinking mode reaches 60.2 on Level-1, while GPT-5 leads Level-2 and Level-3 with 57.5 and 56.4. The same pattern appears among open models. InternVL3-78B is the strongest open model on Levels 1 and 2, yet Intern-S1 is best on Level-3 with 39.6. This instability indicates that local perception, procedural modeling, and experiment-level reasoning are only partially correlated. Reporting only a single overall average would therefore hide meaningful capability differences.

What ExpVid measures, and what it does not. ExpVid is useful because it disentangles three related but distinct abilities: seeing local experimental details, reconstructing procedure, and deriving scientific conclusions. Strong performance on one level does not guarantee strength on the others. At the same time, the benchmark has important limitations. It is built from a single public source, relies on LLM-assisted drafting, does not report inter-annotator agreement, and uses an LLM judge for Level-3. Publicly available videos and papers also leave open the possibility of contamination. These limitations do not negate the benchmark's value, but they define the claims we can make: ExpVid is a targeted measurement instrument for authentic experiment video understanding, not a comprehensive test of scientific intelligence or real-world lab competence.

8 Conclusion

We presented ExpVid, a benchmark for experiment video understanding and reasoning built from 390 authentic JoVE videos and 7,800 verified QA pairs across 10 tasks. The benchmark is organized around a three-level hierarchy spanning fine-grained perception, procedural understanding, and experiment-level scientific reasoning, and its construction emphasizes visual grounding and expert verification.

The benchmark results show that current MLLMs can often recognize local operations and recover coarse order, but they remain much weaker at procedure completion, next-step prediction, and full-experiment reasoning. Visual evidence is indispensable, and explicit thinking mode is not consistently beneficial. We expect ExpVid to be useful

as an evaluation protocol for future work on video-grounded scientific assistants, especially for diagnosing where progress on experimental understanding is genuine and where it is still missing.

References

Anthropic. Claude sonnet 4. `https://www.anthropic.com/claude/sonnet`, 2025. Accessed: 2025-09-21.

Lei Bai, Zhongrui Cai, Maosong Cao, Weihan Cao, Chiyu Chen, Haojiong Chen, Kai Chen, Pengcheng Chen, Ying Chen, Yongkang Chen, et al. Intern-s1: A scientific multimodal foundation model. *arXiv preprint arXiv:2508.15763*, 2025a.

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2. 5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025b.

Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, et al. Are we on the right way for evaluating large vision-language models? *Advances in Neural Information Processing Systems*, 37: 27056–27087, 2024.

Google DeepMind. Gemini 2.5 flash. `https://deepmind.google/models/gemini/flash/`, 2025a. Accessed: 2025-09-24.

Google DeepMind. Gemini 2.5 pro. `https://deepmind.google/technologies/gemini`, 2025b. Accessed: 2025-09-21.

Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 24108–24118, 2025.

Gianmarco Gabrieli, Irina Espejo Morales, Dimitrios Christofidellis, Mara Graziani, Andrea Giovannini, Federico Zipoli, Amol Thakkar, Antonio Foncubierta, Matteo Manica, and Patrick W Ruch. Activity recognition in scientific experimentation using multimodal visual encoding. *Digital Discovery*, 4(2):393–402, 2025.

Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025a.

Dong Guo, Faming Wu, Feida Zhu, Fuxing Leng, Guang Shi, Haobin Chen, Haoqi Fan, Jian Wang, Jianyu Jiang, Jiawei Wang, et al. Seed1. 5-vl technical report. *arXiv preprint arXiv:2505.07062*, 2025b.

Wenyi Hong, Wenmeng Yu, Xiaotao Gu, Guo Wang, Guobing Gan, Haomiao Tang, Jiale Cheng, Ji Qi, Junhui Ji, Lihang Pan, et al. Glm-4.1 v-thinking: Towards versatile multimodal reasoning with scalable reinforcement learning. *arXiv e-prints*, pages arXiv–2507, 2025.

Kairui Hu, Penghao Wu, Fanyi Pu, Wang Xiao, Yuanhan Zhang, Xiang Yue, Bo Li, and Ziwei Liu. Video-mmmu: Evaluating knowledge acquisition from multi-discipline professional videos. *arXiv preprint arXiv:2501.13826*, 2025.

Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, et al. Mvbench: A comprehensive multi-modal video understanding benchmark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22195–22206, 2024.

Ye Liu, Zongyang Ma, Zhongang Qi, Yang Wu, Ying Shan, and Chang W Chen. Et bench: Towards open-ended event-level video-language understanding. *Advances in Neural Information Processing Systems*, 37:32076–32110, 2024.

Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255*, 2023.

Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. *Advances in Neural Information Processing Systems*, 36:46212–46244, 2023.

OpenAI. Gpt-5. `https://openai.com/gpt-5`, 2025. Accessed: 2025-09-21.

Tanik Saikh, Tirthankar Ghosal, Amish Mittal, Asif Ekbal, and Pushpak Bhattacharyya. Scienceqa: A novel resource for question answering on scholarly articles. *International Journal on Digital Libraries*, 23(3):289–301, 2022.

Kimi Team, Angang Du, Bohong Yin, Bowei Xing, Bowen Qu, Bowen Wang, Cheng Chen, Chenlin Zhang, Chenzhuang Du, Chu Wei, et al. Kimi-vl technical report. *arXiv preprint arXiv:2504.07491*, 2025.

Bin Wang, Chao Xu, Xiaomeng Zhao, Linke Ouyang, Fan Wu, Zhiyuan Zhao, Rui Xu, Kaiwen Liu, Yuan Qu, Fukai Shang, et al. Mineru: An open-source solution for precise document content extraction. *arXiv preprint arXiv:2409.18839*, 2024a.

Ke Wang, Junting Pan, Weikang Shi, Zimu Lu, Houxing Ren, Aojun Zhou, Mingjie Zhan, and Hongsheng Li. Measuring multimodal mathematical reasoning with math-vision dataset. *Advances in Neural Information Processing Systems*, 37:95095–95169, 2024b.

Weihan Wang, Zehai He, Wenyi Hong, Yean Cheng, Xiaohan Zhang, Ji Qi, Xiaotao Gu, Shiyu Huang, Bin Xu, Yuxiao Dong, et al. Lvbench: An extreme long video understanding benchmark. *arXiv preprint arXiv:2406.08035*, 2024c.

Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, et al. Internvl3. 5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025.

Xiaoxuan Wang, Ziniu Hu, Pan Lu, Yanqiao Zhu, Jieyu Zhang, Satyen Subramaniam, Arjun R Loomba, Shichang Zhang, Yizhou Sun, and Wei Wang. Scibench: Evaluating college-level scientific problem-solving abilities of large language models. *arXiv preprint arXiv:2307.10635*, 2023.

Takuma Yagi, Misaki Ohashi, Yifei Huang, Ryosuke Furuta, Shungo Adachi, Toutai Mitsuyama, and Yoichi Sato. Finebio: A fine-grained video dataset of biological experiments with hierarchical annotation. *International Journal of Computer Vision*, pages 1–16, 2025.

Jiashuo Yu, Yue Wu, Meng Chu, Zhifei Ren, Zizheng Huang, Pei Chu, Ruijie Zhang, Yinan He, Qirui Li, Songze Li, et al. Vrbench: A benchmark for multi-step reasoning in long narrative videos. *arXiv preprint arXiv:2506.10857*, 2025.

Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9556–9567, 2024.

Yilun Zhao, Haowei Zhang, Lujing Xie, Tongyan Hu, Guo Gan, Yitao Long, Zhiyuan Hu, Weiyuan Chen, Chuhan Li, Zhijian Xu, et al. Mmvu: Measuring expert-level multi-discipline video understanding. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 8475–8489, 2025.

Junjie Zhou, Yan Shu, Bo Zhao, Boya Wu, Shitao Xiao, Xi Yang, Yongping Xiong, Bo Zhang, Tiejun Huang, and Zheng Liu. Mlvu: A comprehensive benchmark for multi-task long video understanding. *arXiv e-prints*, pages arXiv–2406, 2024.

Yuhao Zhou, Yiheng Wang, Xuming He, Ruoyao Xiao, Zhiwei Li, Qiantai Feng, Zijie Guo, Yuejin Yang, Hao Wu, Wenxuan Huang, et al. Scientists' first exam: Probing cognitive abilities of mllm via perception, understanding, and reasoning. *arXiv preprint arXiv:2506.10521*, 2025.